

Matrix inequalities and norms

Lieb's permanental dominance conjecture

Difficulty: extreme **Importance:** interesting to the community**Status:** Partially resolved

Rating rationale: This half-century-old dominance problem unifies many generalized matrix functions; its reach is substantial within matrix and representation theory.

Let $n \geq 1$, let $A = (a_{ij}) \in \mathbb{C}^{n \times n}$ be Hermitian positive semidefinite, let G be any subgroup of the symmetric group S_n , and let χ be the character of any nonzero finite-dimensional complex representation of G . Thus $\chi(e) > 0$, where e is the identity. Define

$$f_\chi(A) = \frac{1}{\chi(e)} \sum_{\sigma \in G} \chi(\sigma) \prod_{i=1}^n a_{i, \sigma(i)}, \quad \text{per } A = \sum_{\sigma \in S_n} \prod_{i=1}^n a_{i, \sigma(i)}.$$

Does the inequality

$$f_\chi(A) \leq \text{per } A$$

hold for every such n, A, G, χ ? The character expression is real for Hermitian A , so the comparison is an ordinary real inequality.

Relevance

Generalized matrix functions include the determinant and immanants. The conjecture seeks a common upper bound across these structured matrix polynomials on the positive-semidefinite cone. It is part of matrix computation and matrix inequality theory, with links to symmetry reductions of tensor powers.

References

- E. H. Lieb, *Proofs of some Conjectures on Permanents*, Journal of Mathematics and Mechanics 16 (1966), 127–134, Conjecture α on p.127 ([publisher's first page](#)).
- I. M. Wanless, *Lieb's permanental dominance conjecture* (2022), Conjecture 1, §2 implication diagram, and §3 partial results ([primary manuscript](#)).
- A. Rico, D. Grinko, R. Krebs, and L. H. Zaw, *Entanglement Structure and Matrix Inequalities from Isotypic Measurements*, Physical Review Letters 137 (2026), 100203; abstract describes immanant inequalities for orders three and four ([journal](#)).
- Y. Li, *Lieb's Permanental Dominance Conjecture for Ordinary Immanants through Order Fifteen*, arXiv:2609.13412v1, submitted September 11, 2026. The primary abstract reports ordinary-immanant dominance through order fifteen and additional infinite families ([version 1 abstract](#)).

Status check — 2026-09-10

Searches for the exact conjecture name and “proof”, “counterexample”, and 2025–2026 found no general resolution. The September 2026 PRL abstract describes fixed-order progress. The August 2026 preprint *GPT, the Counterexample Machine*, §3.1, refutes a stronger conjecture for arbitrary real-stable polynomials; its example is not presented as a PSD-matrix counterexample to Lieb's conjecture ([primary manuscript](#)). Likewise, the older disproof of permanent-on-top does not resolve the displayed statement.

Audit update (2026-09-10): Rechecked Wanless's Conjecture 1 and implication diagram, and the 2026 counterexample paper's discussion. Known immanant families are genuine subcases; the counterexample concerns a stronger real-stable-polynomial assertion, not the displayed PSD statement. This is a bounded literature check, not a proof that no solution exists.

Literature update — 2026-10-06

Li's September 11 preprint reports that every normalized ordinary irreducible immanant is bounded by the permanent for every complex Hermitian positive-semidefinite matrix of order $n \leq 15$. Its abstract identifies the previously remaining partitions $(4,4,3,3)$, $(5,4,3,3)$, and (3^5) , and also reports infinite families including $(m,4,3,3)$ for $m \geq 4$, $(a,b,3,3)$ for $a \geq b \geq 4$ and $5a \geq 8b$, and sufficiently long first rows with fixed tails. These are new ordinary-immanant subcases of the displayed conjecture ([primary abstract](#)).

Scope and remaining gap: The reported results concern irreducible characters of the full symmetric group in the stated orders and families. They do not supply a proof of the displayed statement for all orders, subgroups, and characters. The status remains **Partially resolved**. This update records the primary abstract's scope; no independent audit of the full proof or its finite and formal certificates was performed.